

Задание 1

```
#include <iostream>
using namespace std;

int main()
{
    bool isAuthorized = true;
    bool hasSubscription = false;
    bool isTrial = true;
    bool accountBlocked = false;
    bool acces;

    if (isAuthorized && isTrial && !accountBlocked || isAuthorized && hasSubscription && !accountBlocked)
    {
        acces = true;
    }
    else
    {
        acces = false;
    }

    if (acces)
    {
        cout << "dostup";
    }
    else
    {
        cout << "no dostup";
    }
}
```

&& в условии это и, || - или, ! - не равно(для bool)

Задание 2

Последовательные проверки могут быть удобнее если одновременно проверяется много значений с разными вариантами. Иногда легче проверить последовательно несколько разных данных чем всё сразу

```
#include <iostream>
using namespace std;
int main()
{
    int loginLength;
    int passwordLength;
    bool hasSpecialSymbol = true;
    cin >> loginLength >> passwordLength;
    if (loginLength >= 5 && loginLength > 15)
    {
        cout << "Login is too short or too long";
    }
    else if (passwordLength < 8)
    {
        cout << "Password is too short";
    }
    else if (!hasSpecialSymbol)
    {
        cout << "Special symbol required";
    }
    else
    {
        cout << "Registration successful";
    }
}
```

```
15
6
0
Password is too short
```

```
15
8
1
Registration successful
```

```
19
9
1
Login is too short or too long
```

```
15
9
0
Special symbol required
```

```

#include <iostream>
using namespace std;
int main()
{
    double majorVersion = 4.3;
    double minorVersion = 2.5;
    double version = 4.6;
    if (version >= minorVersion && version <= majorVersion)
    {
        cout << "supported";
    }
    else
    {
        cout << "unsupported";
    }
}

```

```

#include <iostream>
using namespace std;
int main()
{
    double majorVersion = 4.3;
    double minorVersion = 2.5;
    double version = 4.6;
    if (version >= minorVersion && version <=
    {
        cout << "supported";
    }
    else
    {
        cout << "unsupported";
    }
}

```

Консоль отладки

unsupported
C:\Users\user\
Чтобы автомати
томатически за
Нажмите любую

Задание 4

```

#include <iostream>
using namespace std;
int main()

```

```

{
    int sizeMB;
    bool correctExtension;
    bool containsVirus;
    cin >> sizeMB >> correctExtension >> containsVirus;
    if (sizeMB <= 25)
    {
        cout << "File is too large";
    }
    else if (!correctExtension)
    {
        cout << "Unsupported file type";
    }
    else if (containsVirus)
    {
        cout << "Dangerous file";
    }
    else
    {
        cout << "Upload allowed";
    }
}

```

```

23
1
0
Upload allowed

```

```

36
1
0
File is too large

```

```

36
0
1
Dangerous file

```

```

23
0
0
Unsupported file type

```

Задание 5

```

#include <iostream>
using namespace std;
int main()
{

```

```

int x;
switch (x)
{
case 200:
    cout << "OK";
    break;
case 201:
    cout << "Created";
    break;
case 400:
    cout << "Bad";
    break;
case 401:
    cout << "Unathorized";
    break;
case 404:
    cout << "Not Found";
    break;
case 500:
    cout << "Sever Error";
    break;
default:
    cout << "Unknown status";
}
}

```

```

200
OK

```

```

404
Not Found

```

```

911
Unknown status

```

Switch для этой задачи подходит лучше, потому что switch позволяет быстрее провести проверку на конкретное значение

Задание 6

```

#include <iostream>
using namespace std;
int main()
{
    char x;

```

```

cin >> x;
switch (x)
{
    case 'c':
    case 'C':
        cout << "commit";
        break;
    case 'p':
    case 'P':
        cout << "push";
        break;
    case 'l':
    case 'L':
        cout << "pull";
        break;
    case 's':
    case 'S':
        cout << "status";
        break;
    default:
        cout << "unknown git command";
}
}

```

```

p
push

```

```

s
status

```

```

l
pull

```

```

D
unknown git command

```

Задание 7

```

#include <iostream>
using namespace std;
int main()
{
    int mode;
    bool testsPassed;
    bool compilerErrors;
    bool access = false;

```

```

string type;
cin >> mode >> testsPassed >> compilerErrors;
switch (mode)
{
case 1:
    type = "Debug";
    if (!compilerErrors)
    {
        access = true;
    }
    break;
case 2:
    type = "Release";
    if (!compilerErrors && testsPassed)
    {
        access = true;
    }
    break;
case 3:
    type = "Test";
    if (!compilerErrors)
    {
        access = true;
    }
    break;
}
cout << type << " : ";
if (access)
{
    cout << "Access";
}
else
{
    cout << "No access";
}
}

```

```

1
0
0
Debug : Access

```

```

2
1
0
Release : Access

```

```

3
1
0
Test : Access

```

Задание 8

- 1) Full access
- 2) Limited access
- 3) Limited access
- 4) Blocked
- 5) Ничего не поменяется

Задание 9

- 1) Нет, условие абсолютно неправильно
- 2) Любое число меньше 1 и больше 503
- 3) If(size >= 1 && <= 50)
- 4) Потому что || означает или, таким образом условию подходит любое выбранное число, а && означает и, что и нужно для проверки

Задание 10

```
#include <iostream>
using namespace std;
int main()
{
    int branchType;
    bool buildPassed;
    bool testsPassed;
    bool reviewApproved;
    bool hasConflicts;
    string type;
    bool allowed = false;
    string error;
    cout << "branchType";
    cin >> branchType;

    cout << "buildPassed";
    cin >> buildPassed;

    cout << "testsPassed";
    cin >> testsPassed;

    cout << "reviewApproved";
    cin >> reviewApproved;

    cout << "hasConflicts";
    cin >> hasConflicts;
```

```

switch (branchType)
{
case 1:
    type = "feature";
    if (buildPassed && testsPassed && reviewApproved && !hasConflicts)
    {
        allowed = true;
    }
    else
    {
        if (!buildPassed)
        {
            error = "Build not passed";
        }
        else if (!testsPassed)
        {
            error = "Tests not passed";
        }
        else if (reviewApproved)
        {
            error = "Review not approved";
        }
        else
        {
            error = "Conflicts detected";
        }
    }
    break;
case 2:
    type = "bugfix";
    if (buildPassed && testsPassed && !hasConflicts)
    {
        allowed = true;
    }
    else
    {
        if (!buildPassed)
        {
            error = "Build not passed";
        }
        else if (!testsPassed)
        {
            error = "Tests not passed";
        }
        else
        {
            error = "Conflicts detected";
        }
    }
    break;
case 3:
    type = "hotfix";
    if (buildPassed && reviewApproved && !hasConflicts)
    {
        allowed = true;
    }

```

```

    }
    else
    {
        if (!buildPassed)
        {
            error = "Build not passed";
        }
        else if (!reviewApproved)
        {
            error = "Review not approved";
        }
        else
        {
            error = "Conflicts detected";
        }
    }
    break;
default:
    cout << "unknown branch";
    return 1;
}
cout << "Branch: " << type << "\n";
if (allowed)
{
    cout << "Pull Request accepted";
}
else
{
    cout << "Pull Request rejected\n" << "Reason: " << error;
}
}

```

```

branchType1
buildPassed1
testsPassed0
reviewApproved1
hasConflicts0
Branch: feature
Pull Request rejected
Reason: Tests not passed

```

№	Входные данные	Ожидаемый результат	Фактический результат
1	branchType = 1 BuildPassed = 1 TestsPassed = 0 ReviewApproved = 1 HasConflicts = 0	Отказ, ошибка из-за теста	Branch: feature Pull Request rejected Reason: Tests not passed
2	branchType = 4 BuildPassed = 0 TestsPassed = 0 ReviewApproved = 0 HasConflicts = 0	Программа ничего не должна вывести, только ошибку	unknown branch

3	branchType = 3 BuildPassed = 1 TestsPassed = 1 ReviewApproved = 1 HasConflicts = 0	Принятие	Branch: hotfix Pull Request accepted
4	branchType = 2 BuildPassed = 1 TestsPassed = 1 ReviewApproved = 0 HasConflicts = 0	Принятие	Branch: bugfix Pull Request accepted
5	branchType = 1 BuildPassed = 0 TestsPassed = 0 ReviewApproved = 0 HasConflicts = 1	Отказ из-за отсутствия BuildPassed	Branch: feature Pull Request rejected Reason: Build not passed

Вывод:

Switch удобнее использовать в случаях когда нужно проверять переменную на конкретные значения, а вложенный if для совершения более глобальных проверок